

Learning-based optimisation framework for flexibility aggregators in ancillary service markets

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Abstract—This paper addresses the challenge of optimising offering strategies for flexibility aggregators operating in the European manual frequency restoration reserve (mFRR) market. Given the nonlinear and complex behaviour of prosumers, accurately modelling their price responsiveness and flexibility remains a computationally demanding task. To overcome this, we apply an optimisation with a constraint learning (OCL) framework that integrates machine learning surrogate models into optimisation problems. Specifically, we developed a novel linear programming reformulation for rectified linear unit (ReLU) deep neural networks to approximate the responsiveness of prosumers. The framework is applied to model and optimise the profit maximisation problem of the flexibility aggregator, considering rebound effects and cross-temporal constraints. Extensive simulations demonstrate that the proposed non-negative weighted linear programming (NWLP) reformulation significantly outperforms existing methods in terms of both runtime and solution quality, making it a viable approach for modelling nonlinear and nonconvex cost functions in power system optimisation.

Index Terms—constraint learning, neural networks, electricity market, flexibility aggregator, demand response

NOMENCLATURE

Indices and Sets

$\mathbb{N}$	Set of natural numbers
$\mathbb{R}$	Set of real numbers
$\mathbb{R}_+$	Set of non-negative real numbers
$\mathcal{L}$	Index of ReLU DNN layers, $\{1, \dots, L + 1\}$
$\mathcal{T}$	Set of time steps, $\{1, \dots, T\}$

Parameters

λ^M	Market prices DKK/MWh, $\lambda^M \in \mathbb{R}^T$
$\bar{x}$	Maximum obtainable DSF MWh, $\bar{x} \in \mathbb{R}_+^T$
A	Cross-temporal matrix, $A \in \mathbb{R}_+^{T \times T}$
b_l	Bias matrix of layer $l \in \mathcal{L}$, $b_l \in \mathbb{R}^{n_l}$
W_l	Weight matrix of layer $l \in \mathcal{L}$, $W_l \in \mathbb{R}^{n_{l-1} \times n_l}$
L	Number of hidden ReLU layers, $L \in \mathbb{N}$
LB, UB	Bounds for PCTAR reformulation, $LB, UB \in \mathbb{R}$
N_D	Number of training data points, $N_D \in \mathbb{N}$
n_l	Number of neurons in layer $l \in \mathcal{L}$, $n_l \in \mathbb{N}$
N_P	Number of pieces for PWL, $N_P \in \mathbb{N}$
q_t, r_t	Shaping parameters at $t \in \mathcal{T}$, $q_t, r_t \in \mathbb{R}_+$
T	Optimisation horizon, $T \in \mathbb{N}$

Variables

λ^I	Incentive paid for DSF DKK/MWh, $\lambda^I \in \mathbb{R}^T$
λ^P	Total DSF purchase cost DKK, $\lambda^P \in \mathbb{R}_+^T$
$\tilde{x}$	Updated maximum DSF MWh, $\tilde{x} \in \mathbb{R}_+^T$

h_l	Output vector of layer $l \in \mathcal{L}$, $h_l \in \mathbb{R}^{n_l}$
x	DSF reserved for the market MWh, $x \in \mathbb{R}_+^T$

I. INTRODUCTION

To sustain the rapid growth of renewable energy sources, the European Union Agency for the Cooperation of Energy Regulators (ACER) projects that by 2030, the flexibility capacity of the grid will need to double from its current capacity. As conventional generators, once the key to providing supply side flexibility, are being replaced by renewable sources, demand response (DR) has become a critical mechanism to achieve the necessary flexibility of the network [1]. DR mechanisms enable end consumers, including prosumers (those who consume and produce electricity), to adjust their electricity consumption in response to time-varying incentives, whether financial or otherwise.

An extremely popular DR mechanism is the indirect load control of aggregated prosumers. Here, aggregators of behind-the-meter (BTM) flexibility leverage the growing penetration of distributed energy resources (DER), such as rooftop solar panels, electric vehicles, and batteries along with notable advances in home energy management systems (HEMS) to coordinate prosumers in a grid-aware manner. This coordination enables the acquisition of demand-side flexibility (DSF), which can be offered to energy and ancillary services markets. This coordination yields significant benefits: consumers receive financial incentives for shifting their consumption, while system operators gain access to cost-effective and fast-acting DSF. Consequently, state-of-the-art (SOTA) research focuses heavily on aggregator business models and optimal bidding strategies [2].

For aggregators to operate profitably, accurately modelling prosumers' price responsiveness is critical. It is essential for these models to accurately assess the DSF provided by prosumers because system operators apply financial penalties and limit future market participation if promised services are not delivered when needed [3]. In addition, these models enable aggregators to assess various business strategies and engage confidently in the market through optimal bidding. As noted by the authors in [4], prosumers' marginal utility functions (MUFs) or responsiveness functions are often nonlinear, non-smooth, and heavily influenced by uncertainties in electricity prices, load patterns, and weather conditions, making them difficult to model explicitly. These challenges are exacerbated by the lack of real-world data to capture such relationships.

Consequently, many researchers use HEMS models as a proxy for modelling prosumer behaviour within their optimal bidding strategies to represent individual or aggregated prosumers' price responsiveness through nonlinear discomfort/dissatisfaction functions, as demonstrated by the authors in [5] and [6] or by incorporating models of DER and elastic

loads, as done by the authors in [7]. However, these models tend to have scalability issues as the number of prosumers increases, especially if they involve complex optimisation architectures such as bilevel and stochastic optimisation [5].

To address the scalability issue, we previously used an intuitive nonlinear and nonconvex aggregated price responsiveness function in [8], which was then replaced by piecewise linearisation (PWL) to improve tractability. Such aggregated models tend to ignore the effects of uncertain parameters and rebound effect on prosumers' responsiveness functions, as PWL of multivariate functions tends to be computationally inefficient. Thus, there is a pressing need for a scalable surrogate model capable of accurately representing these hard-to-model, nonlinear, nonconvex, and multivariate functions.

An optimisation framework with constraint learning (OCL) employs machine learning-based surrogate models to approximate objective and constraint functions that are difficult to explicitly formulate within optimisation models while ensuring that critical constraints are satisfied [9]. This framework is gaining prominence in modern power system research due to its ability to learn nonlinear, uncertain, or complex interactions in optimisation problems, such as optimal power flow [10] and unit commitment [11]. Furthermore, OCL can dynamically model changes in the stochastic behaviour of parameters with respect to decision variables, as demonstrated in [12], and more accurately capture irrational human behaviour that often influences optimal decision-making [13]. These features position an OCL framework as a promising tool for modelling prosumers' price responsiveness.

The authors in [9] conducted a comprehensive review and identified ReLU-based deep neural networks (DNNs) as one of the most commonly employed machine learning methodologies within the OCL framework. This preference can be attributed to their continuous piecewise linear (CPWL) nature, which enables them to capture a wide range of complex relationships. ReLU DNNs are highly versatile, making them suitable for regression and classification tasks [14]–[16]. Moreover, tight mixed-integer programming (MIP) reformulations exist for incorporating ReLU DNNs into optimisation models, both in the literature and as Python packages [17], [18]. These include the Big-M formulation for max functions, augmented with feasibility-based bound tightening (FBBT) or optimality-based bound tightening (OBBT) algorithms to refine the bounds of ReLU neuron outputs. In FBBT, bounds are tightened by adding cuts to the feasibility space without removing any feasible solutions, whereas in OBBT, bounds are refined by solving a series of optimisation problems. Both methods significantly reduce the search space and improve the computational efficiency of the Big-M formulation [16]. However, MIP reformulations introduce a binary variable for each ReLU neuron, which can substantially increase computational complexity. To maintain tractability, most researchers opt for extremely shallow ReLU DNNs when embedding them into optimisation models [9].

The authors in [19] encountered the challenge of intractability while using ReLU DNN to learn and model battery degradation, where they used the Big-M reformulation of ReLU DNN to schedule multiple batteries. To solve the

intractability issue, the same authors in [20] proposed two linear programming (LP) reformulations named PCAR and PCTAR, which we discuss in more detail in Section III-B. By implementing these modifications, the authors observed an improvement in the tractability of the scheduling problem. However, they pointed out that while PCAR lacked accuracy, PCTAR was accurate and manageable. We note that PCAR and PCTAR are penalty-value-driven methods; hence, their tightness is not guaranteed and depends on the optimisation model, parameter settings, and potentially the ReLU DNN architecture, as demonstrated by the results in Section IV-D.

Therefore, the main identified gap is the lack of a tight LP reformulation for ReLU DNNs that can be used within the OCL framework. In addition, we did not find any OCL-based case studies applied to any BTM flexibility aggregator problem. Hence, to address these gaps, we made the following contributions.

- A tight ReLU DNN LP reformulation to represent cost functions in minimisation problems.
- OCL-based profit maximisation problem for an aggregator of BTM flexibility participating in European manual frequency restoration reserve (mFRR) capacity market.

We note that our proposed reformulation can only represent functions that are minimised. However, it can be a strong candidate for addressing other modern power system problems, including, but not limited to, the minimisation of nonlinear generator cost functions and CO₂ emissions.

The rest of the paper is structured as follows: Section II specifies the mathematical formulations used for the OCL case study. Section III describes various surrogate models used to model prosumer behaviour. Section IV outlines the simulation parameters of this study, presents the subsequent results obtained, and analyses them. Finally, in Section V, we present main conclusions and discuss future work.

II. PROBLEM STATEMENT

In this section, we describe the background information, assumptions, and mathematical models used to develop a profit maximisation problem for a BTM flexibility aggregator.

A. Background and assumptions

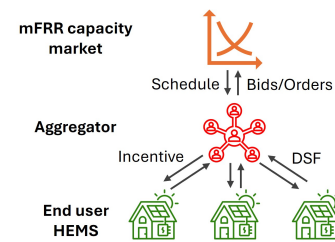


Figure 1: BTM flexibility aggregator business model

Figure 1 illustrates the aggregator business model used in this case study. The aggregator sends an incentive price signal to the HEMS of multiple prosumers. Subsequently, prosumers react by providing a certain amount of DSF. The DSF purchased by the aggregator can then be used to participate in various electricity markets and services. Because each of these markets and services varies significantly depending on

the region, we only consider the European mFRR capacity market operating in Denmark DK1 region to reasonably limit the scope of this case study.

The mFRR capacity market is used to procure tertiary or replacement reserves in both directions to balance demand and supply in cases of emergencies. These markets operate a day before delivery and ensure that reserves are available every hour the next day. These reserves are activated in the event of unexpected emergencies. As a capacity market, all resources acquired through the market process are paid at mFRR capacity market prices, regardless of whether the reserve is activated. Hence, market participants whose bids are accepted must ensure that the accepted reserve is available; otherwise, they are fiscally liable. Interested readers may refer to [3] for more details on the various ancillary markets operated by Energinet and their rules.

To formulate the aggregator's profit maximising problem, we make the following assumptions.

- The aggregator only provides up-regulation reserve to the market, i.e., power flows from prosumer to grid.
- They act as a price-taking entity, and their bids are always accepted. This implies that the aggregator already knows the market-clearing price.
- The aggregator deterministically knows the maximum obtainable DSF from their prosumer cluster at each interval.
- The rebound effect on prosumers' DSF is considered, i.e., DSF activated at an interval will reduce the maximum obtainable DSF in future intervals. For example, current discharge from the prosumer battery forces the battery to charge the lost energy at future intervals. This effect is implemented using cross-temporal weights, which are also assumed to be known deterministically.
- Activations of the mFRR reserves are subject to participation in the mFRR activation market and involve a separate payment if the actual activation occurs. We ignore this market and assume that the reserved capacity is always fully activated at no additional cost to the system operator.
- The prosumer responsiveness function is exactly known.

It is important to note that the assumptions in this study are made to simplify the optimisation model because the primary objective is to evaluate and compare the performance of surrogate models for the prosumer responsiveness function. For a more comprehensive analysis of complex aggregator business models, readers are encouraged to refer to [2] and [21].

B. Mathematical models

Based on the background and assumptions stated in Section II-A, we develop a profit maximising problem for a BTM flexibility aggregator participating in the European mFRR capacity market.

Let $\mathcal{T} = \{1, \dots, T\}$ be the set of time intervals for which the problem is solved with $T \in \mathbb{N}$, $\lambda^M \in \mathbb{R}^T$ be the price profile of the mFRR capacity market in DKK/MWh, $x \in \mathbb{R}^T$ be the DSF reserved for the mFRR market in MWh and $\bar{x} \in \mathbb{R}^T$ be the maximum obtainable DSF from a prosumer group in MWh. Due to the rebound effect, the maximum obtainable DSF is updated to $\tilde{x} \in \mathbb{R}^T$ in MWh. Let $\lambda^I \in \mathbb{R}^T$

be the incentive offered to the prosumer group in DKK/MWh to obtain flexibility.

We define the prosumer responsiveness function based on our previous work in [8], as given in Eq. (1a). This function models the relationship using a saturation curve, as prosumers are typically non-responsive at low prices, with their DSF gradually saturating to the updated maximum achievable DSF, $\tilde{x}_t$, at very high prices. Notably, this approach modifies the methodology proposed in [8] by substituting $\tilde{x}$ for $\bar{x}$. This substitution provides a more realistic representation of the prosumer responsiveness, as the rebound effect appreciates the true value of potential future flexibility. For example, a battery discharged in the current interval requires a greater incentive to discharge in the future intervals. This is because its future flexibility is reduced, making it a more valuable resource.

Therefore, we can see that the prosumers responsiveness function depends on the decision vectors $x, \tilde{x}$, and the saturation curve shaping variables $q_t, r_t \in \mathbb{R}_+$. q_t, r_t are used to represent exogenous factors relevant to this relationship at time $t \in \mathcal{T}$. To avoid bilinear terms within the objective function, we rearrange Eq. (1a) into Eq. (1c), which evaluates the total purchase cost for the reserved DSF $\lambda_t^P \in \mathbb{R}_+$ at time t in DKK. Please note that max function in Eq. (1c) ensures that the incentive paid to prosumers is non-negative. Clearly, this relationship is complex and causes computational difficulties.

$$x_t = \frac{\tilde{x}_t}{1 + e^{(q_t - r_t \lambda_t^I)}} \quad \forall t \in \mathcal{T} \quad (1a)$$

$$\Rightarrow \lambda_t^I = \frac{1}{r_t} \left[q_t - \ln \left(\frac{\tilde{x}_t}{x_t} - 1 \right) \right] \quad \forall t \in \mathcal{T} \quad (1b)$$

$$\therefore \lambda_t^P = \max \left\{ \frac{x_t}{r_t} \left[q_t - \ln \left(\frac{\tilde{x}_t}{x_t} - 1 \right) \right], 0 \right\} \quad \forall t \in \mathcal{T} \quad (1c)$$

Thus, we can define the BTM flexibility aggregator's profit maximisation problem as shown in Eqs. (2a) – (2d).

$$\max_{x, \tilde{x}, \lambda^P} \sum_{t \in \mathcal{T}} [\lambda_t^M x_t - \lambda_t^P] \quad (2a)$$

$$\text{s.t. } x_t \leq \tilde{x}_t \quad \forall t \in \mathcal{T} \quad (2b)$$

$$\tilde{x}_t = \bar{x}_t - \sum_{j \in \mathcal{T}} A_{tj} x_j \quad \forall t \in \mathcal{T} \quad (2c)$$

$$\lambda_t^P = \tilde{f}(x_t, \tilde{x}_t, q_t, r_t) \quad \forall t \in \mathcal{T} \quad (2d)$$

Here, Eq. (2a) sets the objective of the optimisation problem as the maximisation of the net profit of the aggregator. Eq. (2b) ensures that the reserved flexibility is within the updated maximum available DSF. Eq. (2c) accounts for the rebound effect using the cross-temporal matrix $A \in [0, 1]^{T \times T}$, which is a lower triangular matrix with zero diagonal elements. Here, if the DSF is consumed at some interval t , the amount of DSF obtainable at future intervals is reduced. Eq. (2d) replaces the nonlinear and nonconvex cost function λ_t^P with a tractable surrogate model $\tilde{f}(x_t, \tilde{x}_t, q_t, r_t)$. If $\tilde{f}(x_t, \tilde{x}_t, q_t, r_t)$ is based on an ML technique, then the optimisation model is referred to as an OCL-based problem.

III. SURROGATE MODELS FOR λ^P

In this section, we describe the various surrogate models of λ^P used in this study. Here, we consider PWL [22] and ReLU DNN-based approximations to replace the nonlinear function

in Eq. (1c). The reformulations of the ReLU DNN considered include:

- 1) The standard MIP reformulation implemented via Gurobi machine learning (Gurobi ML) Python package [17].
- 2) The LP reformulations presented in [20], namely PCAR and PCTAR, and
- 3) Our proposed reformulation presented in Theorem III.1.

PWL methods are ubiquitous for approximating nonlinear and nonconvex functions in the optimisation literature. However, they suffer the “curse of dimensionality” with both increasing number of inputs and increasing number of pieces [16], making them computationally inefficient for complex relationships. The responsiveness of prosumers depends on a multitude of factors with most likely no closed-form expressions, making PWL impractical for this application. Nonetheless, we treat PWL as a baseline in this study owing to its ubiquity.

Evidently, the purchase cost specified in Eq. (1c) is a multivariate function. Applying the PWL technique to linearise this function will be computationally challenging for a large number of pieces N_P . Hence, we simplify it to a two-dimensional function by assuming that the exogenous factors are deterministically known; that is, q_t, r_t are known beforehand. We then employ the work of the authors in [22] to implement the PWL approximation.

In contrast, ReLU DNNs can easily learn multivariate functions. Hence, we treat the nonlinear function as a four-dimensional function to train the ReLU DNNs. In the rest of this section, we summarise the SOTA and our proposed ReLU DNN reformulations, and the ReLU DNN training strategy employed in this study.

A. Proposed ReLU DNN reformulation

In this section, we discuss the mathematical background, intuition, and key properties of the proposed ReLU DNN LP reformulation.

Definition 1. For any vector $\mathbf{z} \in \mathbb{R}^a$ with any $a \in \mathbb{N}$, we can define the element-wise ReLU function $\sigma : \mathbb{R}^a \rightarrow \mathbb{R}_+^a$, as shown in (3)

$$\sigma(\mathbf{z}) = \begin{bmatrix} \max(z_1, 0) \\ \max(z_2, 0) \\ \vdots \\ \max(z_a, 0) \end{bmatrix} \quad (3)$$

Definition 2. Consider a fully connected ReLU DNN $f : \mathbb{R}^{n_0} \rightarrow \mathbb{R}^{n_{L+1}}$, where $n_0 \in \mathbb{N}$ is the input size and $L \in \mathbb{N}$ is the number of ReLU hidden layers. Let $n_l \in \mathbb{N}$ be the number of output neurons with weights $\mathbf{W}_l \in \mathbb{R}^{n_{l-1} \times n_l}$ and biases $\mathbf{b}_l \in \mathbb{R}^{n_l}$ for each layer $l \in \{1, 2, \dots, L, L+1\}$, where $l = L+1$ is the output layer. Given an input vector $\mathbf{z} \in \mathbb{R}^{n_0}$ such that $\mathbf{h}_0 = \mathbf{z}$, the output of the hidden layers $\mathbf{h}_l \in \mathbb{R}^{n_l}$ and the output of the ReLU DNN are specified in (4a) and (4b), respectively.

$$\mathbf{h}_l = \sigma[\mathbf{W}_l \mathbf{h}_{l-1} + \mathbf{b}_l] \quad \forall l \in \{1, \dots, L\} \quad (4a)$$

$$f(\mathbf{z}) = \mathbf{W}_{L+1} \mathbf{h}_L + \mathbf{b}_{L+1} \quad (4b)$$

Based on the definitions above, we propose Theorem III.1.

Theorem III.1. Let us define functions $c : \mathbb{R}^{n_0} \rightarrow \mathbb{R}$ and $g : \mathbb{R}^{n_0} \rightarrow \mathbb{R}^e$, where $e \in \mathbb{N}$. Let us also consider the ReLU DNN f as defined in (4b). Thus, we can define a cost minimisation optimisation problem of the form below:

$$\min_{\mathbf{z} \in \mathbb{R}^{n_0}} c(\mathbf{z}) + f(\mathbf{z}) \quad (5a)$$

$$\text{s.t. } g(\mathbf{z}) \leq 0 \quad (5b)$$

For some $k \in \mathbb{N}$ and $k \leq L$, if $\mathbf{W}_l \geq 0$ (element-wise) $\forall l \in \{k+1, \dots, L+1\}$, then the optimisation problem can be reformulated as

$$\min_{\mathbf{z} \in \mathbb{R}^{n_0}} c(\mathbf{z}) + f(\mathbf{z}) \quad (6a)$$

$$\text{s.t. } g(\mathbf{z}) \leq 0 \quad (6b)$$

$$f(\mathbf{z}) = \mathbf{W}_{L+1} \mathbf{h}_L + \mathbf{b}_{L+1} \quad (6c)$$

$$\mathbf{h}_0 = \mathbf{z} \quad (6d)$$

$$\mathbf{h}_l = \sigma[\mathbf{W}_l \mathbf{h}_{l-1} + \mathbf{b}_l] \quad \forall l \in \{1, \dots, k-1\} \quad (6e)$$

$$\mathbf{h}_l \geq \mathbf{W}_l \mathbf{h}_{l-1} + \mathbf{b}_l \quad \forall l \in \{k, \dots, L\} \quad (6f)$$

$$\mathbf{h}_l \geq 0 \quad \forall l \in \{k, \dots, L\} \quad (6g)$$

Proof. Let us define a simplified version of the problem, that is, a ReLU DNN with a single neuron:

$$\min_{z \in \mathbb{R}} f(z) \quad (7a)$$

$$\text{s.t. } g(z) \leq 0 \quad (7b)$$

$$f(z) = \max(az, 0) \quad (7c)$$

The standard LP reformulation of this problem is

$$\min_{z \in \mathbb{R}} f(z) \quad (8a)$$

$$\text{s.t. } g(z) \leq 0 \quad (8b)$$

$$f(z) \geq az, f(z) \geq 0 \quad (8c)$$

Because the problem aims to minimise $f(z)$, this reformulation will ensure $f^*(z) = az$ or $f^*(z) = 0$. If it were a maximisation problem, $f(z)$ would be unbounded. Thus, replacing max (or ReLU) constraints with linear inequalities requires the output of the neuron to be minimised.

Now let us consider a problem with nested maximum constraints, such as a ReLU DNN:

$$\min_{\mathbf{z}, \mathbf{u} \in \mathbb{R}^2} f(\mathbf{z}) \quad (9a)$$

$$\text{s.t. } g(\mathbf{z}, \mathbf{u}) \leq 0 \quad (9b)$$

$$f(\mathbf{z}) = \max \left(\sum_{j=1}^2 a_j z_j, 0 \right) \quad (9c)$$

$$z_i = \max \left(\sum_{j=1}^2 b_{ij} u_j, 0 \right) \quad \forall i \in \{1, 2\} \quad (9d)$$

Here, $a \in \mathbb{R}^{2 \times 1}$ and $b \in \mathbb{R}^{2 \times 2}$ are the weight matrices of the ReLU DNN layers. A reformulation similar to the one demonstrated above would look as follows:

$$\min_{\mathbf{z}, \mathbf{u} \in \mathbb{R}^2} f(\mathbf{z}) \quad (10a)$$

$$\text{s.t. } g(\mathbf{z}, \mathbf{u}) \leq 0 \quad (10b)$$

$$f(\mathbf{z}) \geq \sum_{j=1}^2 a_j z_j, \quad f(\mathbf{z}) \geq 0 \quad (10c)$$

$$z_i \geq \sum_{j=1}^2 b_{ij} u_j, \quad z_i \geq 0 \quad \forall i \in \{1, 2\} \quad (10d)$$

If $a_1, a_2 \geq 0$, the optimisation problem aims to minimise z_1 and z_2 binding the max constraint. However, if $a_1 < 0$ or $a_2 < 0$, the problem aims to maximise z_1 or z_2 , respectively. Thus, the associated max constraint becomes unbounded. Solving this problem will provide solutions different from the original problem. Applying this argument, we justify the reformulation presented in (6). Consequently, a ReLU DNN can be replaced by the proposed LP reformulation within optimisation problems if it satisfies two conditions: 1) the output of the ReLU DNN is minimised and 2) the weight matrix of all layers barring the first ReLU layer must be non-negative matrices. $\square$

Definition 3. A ReLU DNN f , as defined in Definition 2, is called a non-negative weighted (NW) ReLU DNN if all weight matrices are non-negative from the second layer onwards, i.e., $\mathbf{W}_l \geq 0$ (element-wise) $\forall l \in \{2, \dots, L+1\}$.

Clearly, an NW ReLU DNN can be equivalently represented through a LP reformulation, leveraging the results established in Theorem III.1. Henceforth, we refer to this formulation as the NWLP reformulation. While NW ReLU DNNs have been extensively studied in the literature, particularly for their monotonicity properties, we are the first to propose an LP reformulation for these networks and demonstrate its application within the OCL framework.

ReLU DNNs are universal approximators, but in applications like finance and healthcare, preserving monotonicity is crucial. One way to achieve this is to constrain the sign of the weights in ReLU DNNs. However, these constraints affect the expressiveness of ReLU DNNs, limiting them to learning and accurately representing only convex and monotonic functions, as demonstrated in [23]. NW ReLU DNNs are restricted to exclusively convex and monotonically increasing functions. This limitation poses a caveat when applying this formulation, as it trades accuracy and expressiveness for tractability and tightness, as demonstrated by our results in Section IV. Despite this limitation, the results in Section IV demonstrate that NW ReLU DNNs can effectively capture nonlinear and nonconvex monotonically increasing flexibility cost functions for aggregator operation. Consequently, NW ReLU DNNs may also be suitable for learning nonlinear fuel cost functions or CO₂ emissions, which are inherently increasing monotonically.

B. SOTA ReLU DNN reformulations

The authors in [20] proposed two LP reformulations for ReLU activation functions: PCAR and PCTAR. In PCTAR, the ReLU function is approximated by a triangle setting the relevant upper and lower bounds (UB, LB) on the input to the neuron, as shown in Fig. 2a, whereas in PCAR, the entire input space is considered, as shown in Fig. 2b.

From Fig. 2, it is clear that they both apply the same principle as our proposed NWLP reformulation, which is minimisation of the max function. However, they used the

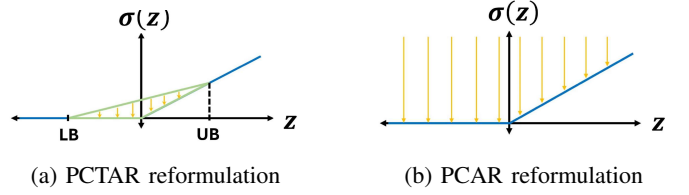


Figure 2: ReLU function LP reformulations adapted from [20]

penalty function mechanism to minimise the output of each ReLU neuron. Thus, they offer greater generalisability and can serve as LP reformulations applicable to all ReLU DNNs, unlike the limitations of our proposed NWLP reformulation. However, the tightness of PCAR and PCTAR reformulations is not guaranteed and depends on whether the penalty costs are sufficiently chosen. This is not the case for the proposed NWLP reformulation, which guarantees tightness for NW ReLU DNNs whose output is minimised, as discussed in Section III-A.

C. ReLU DNN training strategy

In this section, we describe the training strategy for ReLU DNNs to learn the nonlinear and non-convex relationship presented in Eq. (1c). We note here that despite its non-convex nature, our aim is to demonstrate that NW ReLU DNNs may sufficiently capture the relationship to obtain reasonable quality solutions. The defined function has four inputs and one output, and is invalid if $x_t \geq \tilde{x}_t$. It should be noted that two sets of ReLU DNNs are trained using the strategy described below: 1) **NW ReLU DNN**, which is implemented using the NWLP formulation as specified in Section III-A; 2) **Unconstrained (UC) ReLU DNN**, which is implemented using the other reformulations.

Let $\mathcal{D} \in \mathbb{R}_+^{N \times 4}$ denote input dataset with $N_D \in \mathbb{N}$ data points. These points are randomly generated such that an input point $\mathbf{z} = \{x_t, \tilde{x}_t, q_t, r_t\}$ satisfies $\underline{\mathbf{z}} \leq \mathbf{z} \leq \bar{\mathbf{z}}$ and $z_1 < z_2$. Here, $\underline{\mathbf{z}}$ and $\bar{\mathbf{z}}$ are the lower and upper bounds of $\mathbf{z}$. These are then substituted in Eq. 1c, to obtain the corresponding output dataset $\mathcal{O} \in \mathbb{R}_+^N$, and so the combined dataset is denoted as $\{\mathcal{D}, \mathcal{O}\}$.

To expedite the training process, we applied min-max normalisation to the combined dataset, normalising the input and output values to a range of $[0, 1]$. The resulting normalised combined input and output datasets, denoted as $\{\mathcal{D}, \mathcal{O}\}'$, were then randomly divided into training and validation sets in an 80:20 ratio. We trained the ReLU DNNs using the Adam optimiser with a learning rate of 0.0001 while applying the mean squared error (MSE) loss metric. The models were trained for 1000 epochs with a batch size of 1000. The validation set was used to monitor the training process and prevent over-fitting.

IV. SIMULATION STUDY AND RESULTS

In this section, we describe the simulation studies to evaluate the performance of the proposed reformulation in Section II-B compared to SOTA methods.

A. Optimisation parameters

In this section, we describe the market and prosumer cluster parameters used within the optimisation model. As described

in Section II-A, we assume that the aggregator participates in the European mFRR capacity market, specifically in the DK1 region of Denmark, which is operated by Energinet, the Danish transmission system operator. This market is operated one day in advance and reserves tertiary FCAS for every hour the next day. Thus, the optimisation horizon is $T = 24$ hours.

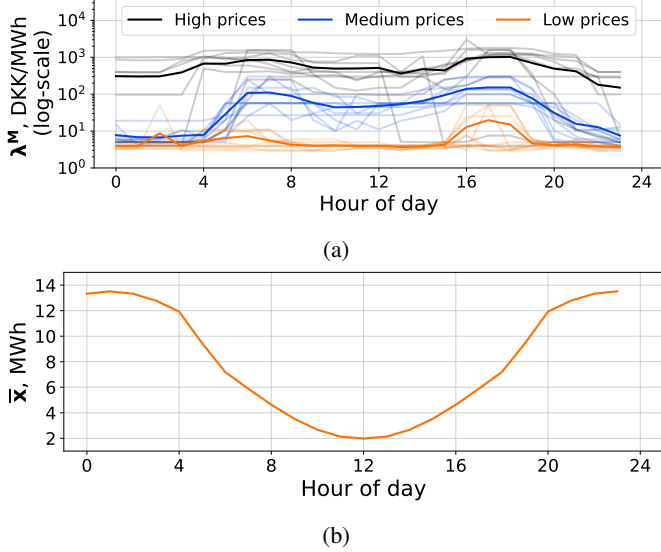


Figure 3: Optimisation parameters: (a) mFRR capacity market price by category; (b) Maximum DSF obtainable

For this study, we obtained real market price scenarios for mFRR capacity from the API of the Energinet data service [24]. We separated the price scenarios into low, medium, and high prices and selected ten scenarios in each of the above categories. This was done to study the influence of market prices on the tractability and performance of the approximation methods of λ^P . The price profiles for these categories are illustrated in Fig. 3(a) with the mean price profile highlighted. We would like to note that 90% of the prices are below 10 DKK/MWh, making the low-price scenarios highly likely and consequently the critical case. However, because of the nature of this market, when prices are high, they are orders of magnitude higher, as demonstrated by the log-y scale in Fig. 3(a). Therefore, studying the performance of the reformulations for these three price categories is crucial for this application.

The data of the prosumer cluster was generated randomly based on our previous work [8]. The maximum DSF obtainable profile ($\bar{x}$) used in this study is shown in Fig. 3(b). Note that this profile follows a sinusoidal pattern to account for diurnal factors, as demonstrated by the work done in [25]. The cross-temporal matrix (A) is a randomly generated lower triangular matrix with diagonal elements equal to zero. The price elasticity of prosumers can be used as a proxy for A , which can be obtained using real-world data, as shown by [26]. Furthermore, the total DSF lost in future intervals must be less than or equal to the DSF reserved in the current interval t , that is, $\sum_{i \in \mathcal{T}} A_{i,t} \leq 1, \forall t \in \mathcal{T}$.

For the prosumer responsiveness model, the shaping parameters q_t, r_t were randomly generated and fixed for each hour of the optimisation horizon. These parameters can be

derived by fitting a saturation curve to real-world data using regression [27]. For the two-dimensional PWL model [22], we set $N_P = 4$ four for both inputs.

For the PCTAR method, bounds for the neurons were set as $LB = -10, UB = 10$ to account for potential outliers, given that training with min-max normalisation typically keeps neuron inputs and outputs within ± 1 . To train the ReLU DNNs, $N_D = 3 \times 10^5$ data points were generated following the methodology in Section III-C. We note that the average CPU training time for all the ReLU DNN architectures used in this case study was 587 seconds.

B. Tractability and accuracy analysis

In this section, we study the tractability and accuracy of the λ^P approximation methods for the OCL case study described in Section II-B. For this, we trained UC and NW ReLU DNNs with three hidden layers consisting of 10, 20, and 10 neurons, respectively. These trained neural networks are used to represent $\lambda_t^P, \forall t \in \mathcal{T}$ within our optimisation model using the Gurobi ML package, PCTAR, PCAR, and NWLP reformulations. In addition, we employed the PWL method as a benchmark for evaluating these approaches based on neural networks. The formulated optimisation models were then solved using Gurobi[®] 11.0 [28] on a PC with 16GB RAM and an Intel[®] i7 16-core processor. We set the stopping criteria to be 3600 seconds or MIP Gap of 1%, whichever came first. The results are summarised in Table I. Here, the realised profit is calculated by substituting the optimal decision vectors $x^*, \tilde{x}^*$ into Eqs. (1c) and (2a). Since PCAR and PCTAR are penalty value driven approaches, we tried multiple penalty values while running the optimisation, as explained in Section IV-D. For comparison in this section, we chose the penalty value that maximised realised profit and minimised runtime for each individual price scenario.

Table I: Tractability analysis under different price scenarios

Price case	Surrogate model	Mean realised profit DKK	Mean run-time	Mean MIP gap %
High	PWL	50402	1 s	1
	Gurobi ML	50253	3 s	1
	PCTAR	50861	44 ms	0
	PCAR	50861	26 ms	0
	NWLP	50670 (0.4%)	34 ms	0
Medium	PWL	3910	6 s	1
	Gurobi ML	3678	8 s	1
	PCTAR	3870	44 ms	0
	PCAR	3870	31 ms	0
	NWLP	3894 (0.4%)	31 ms	0
Low	PWL	229	29 s	0
	Gurobi ML	171	15 s	1
	PCTAR	-514	65 ms	0
	PCAR	-514	48 ms	0
	NWLP	226 (1.3%)	35 ms	0

Figure 4(a) shows a histogram of the CPU runtime for all price scenarios solved using various surrogate models. CPU runtimes provide insight into the amount of computer resources being used, which in turn affects the operational

costs for the aggregator. Please note that the x-axis of this plot is on a log scale. As expected, we can see that the LP reformulations PCTAR, PCAR, and NWLP significantly outperform the PWL and Gurobi ML reformulations in terms of speed and CPU resource usage, by a factor ranging from hundreds to thousands. This is because parallel processing is used by Gurobi® for branch-and-bound algorithms to solve MIPs. Note that CPU times reported here are only indicative in nature as Python modules cannot provide the exact CPU runtime; however, they are proportional.

From Table I, we note that all models are relatively fast in the high price scenarios, followed by medium price scenarios. This can be explained in Fig. 3(a), where the high price scenarios are orders of magnitude larger and more volatile than the low price cases and the purchase cost of prosumer flexibility. Thus, the decision for the optimiser is fairly straightforward, as it is easier for the branch-and-bound algorithm to reject sub-optimal solutions. Conversely, in the case of low-price scenarios, the purchase cost and mFRR prices are most likely comparable making it harder for the optimiser to identify and reject sub-optimal solutions.

From Table I, we can also see that for the high price scenarios, PCTAR and PCAR provide the highest mean realised profit and for the other two price scenarios, PWL provides the best realised profit. In all three cases, our proposed formulation is a close second in terms of mean realised profit, as shown by the percentage difference values in parentheses. Interestingly, PCAR and PCTAR approaches, on average, provide losses in the low-price case scenarios, which is the most likely and critical case scenario. This is because these methods tend to underestimate the purchase cost of DSF, as evidenced by the elevated RMSE values depicted in Figure. 4(b), which illustrates the realised RMSE for every approximation technique across the range of price scenarios. This underestimation is effective when dealing with high- or medium-price situations, but fails in low-price scenarios. In sharp contrast, our NWLP method achieves relatively lower RMSE despite being based on the NW ReLU DNN, which has lower expressiveness for non-convex functions when compared to the UC ReLU DNN. The reason for this is explored further in Section IV-C.

Figure 4(b) also shows that the PWL approximation achieves the lowest RMSE for the three price scenarios. However, our NWLP reformulation achieves approximately the same quality of solutions with a much higher computational efficiency. Additionally, PWL methods are impractical in real-world situations because the purchase cost of DSF would most likely be modelled by a multi-input data-driven approach. Therefore, the proposed NWLP method is arguably a more practical option to represent such cost functions and achieve a high level of accuracy, tractability, and solution quality for this application.

C. ReLU DNN length and width sensitivity analysis

In this section, our objective is to study the tractability and accuracy performance of ReLU DNN-based approximation methods with varying architectures, that is, the number of neurons (width) and hidden ReLU layers (length). Here, two sets of cases were considered: 1) varying the number of widths

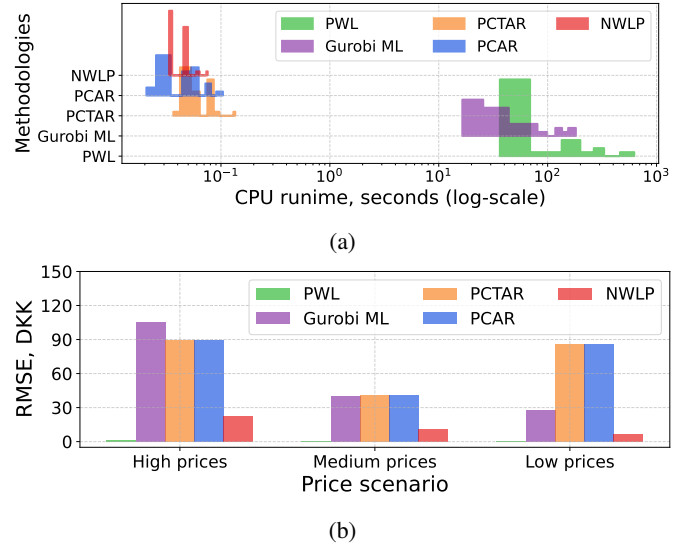


Figure 4: Tractability and accuracy analysis: a) CPU runtime histogram; b) Realised RMSE for various price scenarios

with a fixed length and 2) varying the lengths with a fixed width. Here, we only considered the ten low-price scenarios, because they represent the most likely and critical cases.

Recall from Section IV-B, the ReLU DNN had a length of three with 10, 20 and 10 neurons, respectively, that is, a total width of 40. As noted in section IV-B, for each architecture specified here, we train two sets of ReLU DNNs: UC for implementation via Gurobi ML, PCTAR, and PCAR reformulations, and NW for implementation via the NWLP reformulation. In addition, we ran a penalty sensitivity analysis for PCAR and PCTAR for all ReLU DNN architectures, as discussed in Section IV-D. We then choose the best penalty values that maximise the realised profits and minimise the runtime for each case reported in this section.

For the width sensitivity analysis, we set the length to three and varied the widths to 20, 80, 200, and 400. Note that the neurons are distributed in proportion to the three layers of the ReLU DNN mentioned above.

In Table II, we report the mean realised profit, runtime, and MIP gap in % for feasible solutions of the width sensitivity analysis. Clearly, the runtime increased as the width of the ReLU DNN increased. This is expected because the number of constraints is linked to the number of neurons. The NWLP method provides the best mean realised profit for all cases and is computationally efficient in doing so. Even in one case where it is slower than PCTAR, it is only about 0.8 milliseconds slower while realising significantly higher profits. Gurobi ML model provides sub-optimal solutions with increasing width, as demonstrated by the mean MIP gap. Please note that for the width of 400 neurons, 8 out of 10 price scenarios were not feasibly solved within 3600 seconds using the Gurobi ML method. Thus, the mean MIP gap reported is only for the two scenarios that could be solved as indicated by *. This shows that Gurobi-ML models cannot tractably represent wider neural networks, which may create issues in applications requiring wider ReLU DNNs.

Figure 5(a) shows the realised RMSE for all price scenarios for each width case. We also reported the training RMSE

Table II: Width sensitivity analysis

Width	Surrogate model	Mean realised profit DKK	Mean run-time	Mean MIP gap %
20	Gurobi ML	227	2 s	0
	PCAR	-214	15 ms	0
	PCTAR	-214	8 ms	0
	NWLP	231	16 ms	0
80	Gurobi ML	221	3600 s	127
	PCAR	-36	78 ms	0
	PCTAR	-36	181 ms	0
	NWLP	237	56 ms	0
200	Gurobi ML	115	3600 s	884
	PCAR	-147	549 ms	0
	PCTAR	-147	921 ms	0
	NWLP	239	259 ms	0
400	Gurobi ML	-18	3601 s	236*
	PCAR	-9	3 s	0
	PCTAR	-22	6 s	0
	NWLP	238	748 ms	0

for the ReLU DNN for each width case. It can be seen that for the NWLP reformulation, RMSE is close to the training RMSE of the ReLU DNN it approximates. This demonstrates the tightness of our formulation. In sharp contrast, the PCAR and PCTAR approximations have a higher RMSE than the ReLU DNN. This is because penalty value driven methods do not guarantee tightness of formulation in all cases. This also explains why the mean realised profit is negative for these cases, since these models may allow for solutions where the purchase cost of DSF is severely underestimated when compared to their true values. Surprisingly, the Gurobi ML model also displays behaviour similar to PCAR and PCTAR for widths of 80 and 200. This indicates that their formulations are not tight. However, this may be due to the 3600-second time limit, as the OBBT algorithms they use may relax the problem to obtain candidate solutions and then tighten it over multiple iterations [17], [18].

We also noted that the RMSE of the trained UC ReLU DNN decreased with increasing width. This is because the expressiveness of a DNN increases with the number of neurons [15]. However, NW ReLU DNN RMSE remains flat with increasing width. As discussed in Section III-A, NW ReLU DNNs are expected to have limited expressiveness capabilities owing to their NW constraints. This is a limitation of NWLP reformulation; nonetheless, it seems to outperform the other methods for this application.

For the length sensitivity analysis, we used the following architectures: 1) length of one with 40 neurons; 2) length of two with 20 and 20 neurons, respectively; 3) length of four with 5, 15, 15, 5 neurons, respectively; and 4) length of five with 2, 8, 20, 8, 2 neurons, respectively. Note that the total width is fixed to 40 for all cases.

In Table III, we report the mean profit realised, the runtime, and the MIP gap % for the length sensitivity analysis. The length appears to have little effect on the mean runtime of the PCAR, PCTAR, and NWLP reformulations, remaining roughly the same across different lengths. However, increasing length increases the mean runtime in Gurobi ML, except for

Table III: Length sensitivity analysis

Length	Surrogate model	Mean realised profit DKK	Mean run-time	Mean MIP gap %
1	Gurobi ML	234	4 s	1
	PCAR	-359	13 ms	0
	PCTAR	-359	27 ms	0
	NWLP	239	18 ms	0
2	Gurobi ML	233	87 s	1
	PCAR	-118	38 ms	0
	PCTAR	-118	70 ms	0
	NWLP	237	30 ms	0
4	Gurobi ML	209	53 s	1
	PCAR	106	37 ms	0
	PCTAR	106	59 ms	0
	NWLP	234	27 ms	0
5	Gurobi ML	156	3600 s	55
	PCAR	47	12 ms	0
	PCTAR	47	32 ms	0
	NWLP	219	35 ms	0

length 4 where it decreased. In addition, the runtime suddenly jumps to 3600 seconds for length 5, making it difficult to comment on the tractability behaviour of Gurobi ML model for deeper neural networks. Please note that the exact Gurobi ML algorithm is not publicly available, so it is difficult to find a reason for this behaviour.

The PCAR and PCTAR methods are able to provide profitable results when the length is greater than 4. Thus, there are some relationships between the length of the ReLU DNN, the penalty values, and the performance of the PCAR and PCTAR formulations. This indicates that using these approximations may not be straightforward. Depending on the function they approximate, different architectures of ReLU DNNs and penalty values must be tested to determine the optimal hyperparameters. Conversely, the NWLP method provides the best mean realised profit while being one of the most computationally efficient for all lengths.

Figure 5(b) shows the realised RMSE for all price scenarios and the training RMSE for both UC and NW ReLU DNNs for each length case. We note that, similar to the width case, the realised RMSE of the NWLP method is close to the training RMSE of the ReLU DNN, further demonstrating the tightness of our methodology. Similar to the varying width case, the NW RMSE remained flat because of the lower expressiveness of the NW ReLU DNNs. In addition, PCAR and PCTAR RMSE are significantly higher than UC training NN RMSE, again confirming that this reformulation is not a tight approximation of ReLU DNNs. Interestingly, higher-length UC ReLU DNNs have a higher training RMSE, most likely due to the unnecessary addition of hidden layers to the model and/or improper batch-size or epoch values. This is also reflected in the mean realised profit by the Gurobi ML approach, which is reduced for lengths 4 and 5.

In addition, Gurobi ML stops being a tight reformulation at lengths 4 and 5. For length 5, the reasoning is the same as in the case of varying widths, which is the time limit for solving the problem. However, in the case of length 4, the problem is solved below the 1% MIP gap within the time

limit and still displays a higher RMSE. Upon closer inspection, we identified that two of the ten price scenarios skewed the results. Removing these scenarios provided an RMSE of 1.955 DKK, which was very close to the training RMSE. We also note that the Gurobi ML approach has tractability issues with both increasing length and width of ReLU DNNs, and so may not be an appropriate surrogate for deeper and complex ReLU DNNs.

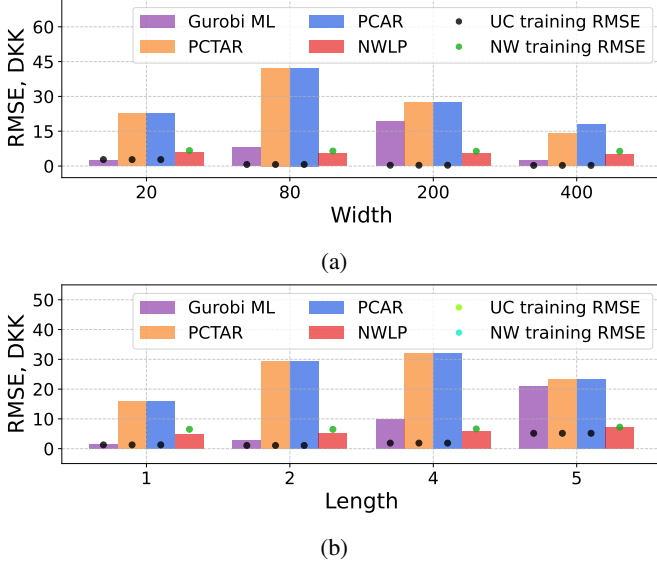


Figure 5: Accuracy sensitivity analysis: a) ReLU DNN with varying widths; b) ReLU DNN with varying lengths

In conclusion, these results demonstrate that the NWLP method is a tight reformulation of NW ReLU DNNs, irrespective of the architecture and optimisation parameters. We observe that our approach offers high-quality solutions at notably rapid speeds, even though it shows a diminished ability to learn nonconvex functions, as reflected by the substantially higher training RMSE. Hence, the proposed approach may be a good candidate for representing a nonlinear and even nonconvex monotonically increasing cost functions, depending on the application.

D. Penalty sensitivity analysis for PCAR and PCTAR

Because PCTAR and PCAR are penalty-driven approaches, we performed a penalty sensitivity analysis to determine the best penalty value for each price scenario and the ReLU DNN architecture. We considered two classes of penalty values: 1) constant penalty values applied to all neurons 0.01, 1, 10, 1000 and 2) layer-based penalty values $5^l, 2^l, 2^{-l}, 5^{-l}, 10^{-l}$ for the l^{th} hidden ReLU layer.

Figure 6 shows sample penalty sensitivity results for the three sets of price scenarios. The x-axis specifies the realised profit and the y-axis specifies the runtime for the scenario. The best penalty value is one which maximises realised profit and minimises runtime; hence, values closer to the bottom right corner of the graph. From the three examples, we can see that the best penalty value may change across cases and the profit obtained varies significantly with the penalty value chosen.

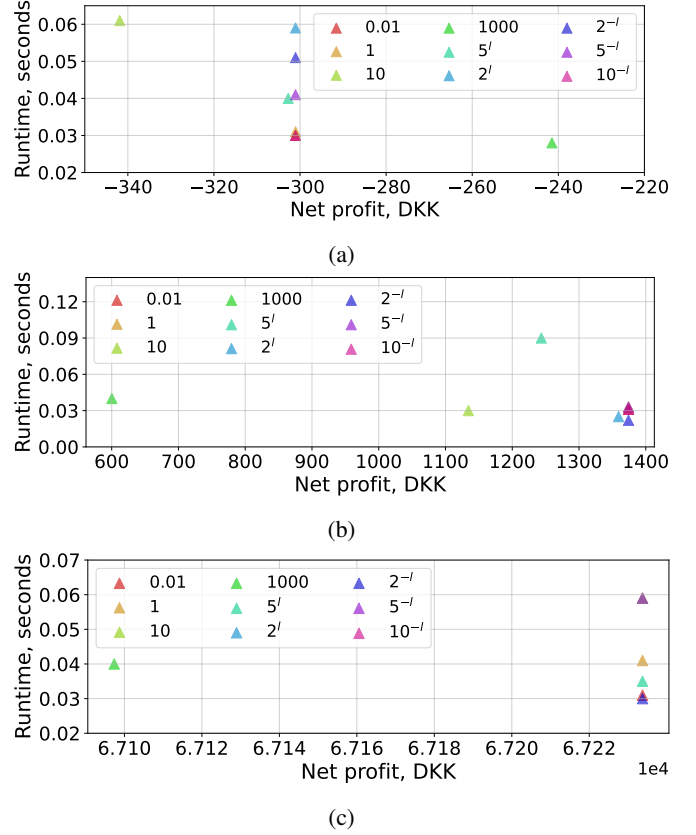


Figure 6: Penalty sensitivity analysis examples: a) Low price scenario; b) Medium price scenario; c) High price scenario

Furthermore, we identified the optimal penalty value across all cases for the high-price, medium-price, and low-price scenarios: $5^{-l}, 2^{-l}, 1000$ for the PCTAR reformulation and 1, $5^{-l}, 1000$ for the PCAR reformulation based on the study conducted in Section IV-B. We also observed that, regardless of the variations in length and width discussed in Section IV-C, 1000 was the best penalty value for both the PCAR and PCTAR formulations. This suggests that the penalty value may be strongly correlated with the optimisation parameters. However, in other applications where the optimisation parameters are more evenly distributed, identifying the best penalty value may be more challenging. Also, the RMSE of PCAR and PCTAR are generally equal, except when the width is set to 400, where PCAR performs better. This discrepancy is due to insufficient LB, UB selection in PCTAR, leading to worse profit, as shown in Table II, further emphasizing the complexity of using PCTAR.

Although the PCAR and PCTAR methods are generally applicable to all classes of optimisation problems, the results from this section and Section IV-C indicate that different combinations of ReLU DNN architectures, and penalty and LB, UB values must be tested to achieve satisfactory performance. Even when satisfactory results are achieved, it cannot be guaranteed that the same penalty values will consistently perform well across all optimisation parameters. In contrast, the proposed NWLP approach is much easier to implement and can guarantee accuracy comparable to that of the NW ReLU DNN it approximates.

V. CONCLUSIONS AND FUTURE SCOPE

In this paper, we proposed an LP reformulation for ReLU DNNs, called NWLP reformulation, which is theoretically designed to represent nonlinear, convex and monotonically increasing cost functions minimised within an optimisation problem. We applied this approach to an OCL-based profit maximisation problem for an aggregator of BTM flexibility participating in the European mFRR capacity market. In this application, the cost of BTM flexibility was modelled as a nonlinear, nonconvex function and approximated using PWL, SOTA ReLU DNN reformulations, and the NWLP reformulation. The results demonstrated that the NWLP method provided the most computationally efficient solutions while maintaining high quality. Furthermore, the NWLP ReLU reformulation exhibited tightness across varying ReLU DNN architectures, making NW ReLU DNNs a promising surrogate model for prosumers' responsiveness functions within an OCL framework, particularly as more data on prosumer behaviour becomes available.

Despite these benefits, NW ReLU DNNs have certain limitations, including reduced expressiveness, lower accuracy, and the theoretical ability to represent only nonlinear and convex monotonically increasing functions. However, our results indicate that nonconvex cost functions can be effectively represented within optimisation models using this reformulation. Therefore, the NWLP reformulation warrants further exploration as a viable candidate for representing complex, and even nonconvex cost functions, such as prosumers' responses to price signals, CO₂ emissions, or nonlinear generator cost functions in modern power system optimisation problems.

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